

Which Wireless Technology Should You Use?

No wireless technology offers everything that you want for an application. You have to choose one over the other, and then work on it to get your product working in just the way you want it to. This article helps you make the right choice

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There are a large number of wireless technologies available, both standardised and proprietary. Having assessed the most used technologies (to ensure a decent level of compatibility) in that list, this article presents you with the best suited applications for each technology, as well as the alternatives that are available for it.

In order to weed out the incompatibility issues linked to proprietary wireless technologies over the long run, we won't consider them too much in this article, but will give more weight to the standardised wireless technologies.

ZigBee

If you are developing a wireless sensor network (WSN), industrial automation project, home entertainment and control system, or a building automation system, ZigBee performs the best.

ZigBee (IEEE 802.15.4) is known for its low power consumption and inherently long battery life. It is also known for its low duty cycle that allows ZigBee-powered sensors to remain in limbo for extended periods if the use-case permits. Additionally, by utilising high-density mesh networks, it allows low latency and also enables the creation of a self-healing network. The designer is also able to tap into a decent data rate, and speed ranges from 20 kbits/s in the 868MHz band (Europe) to 250 kbits/s in the 2.4GHz band.

The best fit. ZigBee is the best fit for applications that typically require



low power and long life for battery-powered sensors, a low data rate and a secure network. ZigBee also provides a line-of-sight range of up to 80 metres. When one central node is not reliable one can also use ZigBee to implement mesh networks, which use intermediate nodes to transfer data over long distances and are able to find new paths if one node goes offline.

"ZigBee is the only standards-based wireless technology designed to address the unique needs of low-cost, low-power wireless sensor and control networks in just about any market. Since 2.4GHz ZigBee can be used almost anywhere, is easy to implement and needs little power to operate, the opportunity for growth into new markets, as well as innovation in existing markets, is limitless," says Oliver Smith, director sales, Telegesis UK Ltd.

Alternatives.

1. *Need a higher data rate?* If the use-

case requires higher data rates than what ZigBee provides, *Bluetooth Smart* is the technology to go for.

2. *Have Wi-Fi working in the same area?* Since Wi-Fi operates in the same band as ZigBee, the former has a tendency to affect ZigBee's operation and increase latency. It does not affect deliverability, but slows down the process. In this case, you can try out Z-Wave.

Bluetooth Smart

With the advent of IPv6 and the drive to connect the next billion devices to the Internet, we have seen tremendous growth in the use of sensors. The extremely small and efficient sensors available today help multiple applications to get 'smarter.' Bluetooth Smart allows a large number of communication nodes to connect together with limited latency requirements, making it an ideal solution for this sort of im-